Digital Certificates, Modern Public Key Infrastructure (PKI) with Ed25519

Final Certificate:

```
PS C:\Users\shevc> cat server.crt
-----BEGIN CERTIFICATE-----
MIIBvzCCAXGgAwIBAgIUa3fs7YUkVOTJBwuY7i0cfvlRaXwwBQYDK2VwMFsxCzAJ
BgNVBAYTAlVBMRMwEQYDVQQIDApTb21lLVN0YXRlMSEwHwYDVQQKDBhJbnRlcm5l
dCBXaWRnaXRzIFB0eSBMdGQxFDASBgNVBAMMC0xhYiBSb290IENBMB4XDTI2MDQw
NjE1MzE0MFoXDTI3MDQwNjE1MzE0MFowYDELMAkGA1UEBhMCQVUxEzARBgNVBAgM
ClNvbWUtU3RhdGUxITAfBgNVBAoMGEludGVybmV0IFdpZGdpdHMgUHR5IEx0ZDEZ
MBcGA1UEAwwQc2VjdXJlLWxhYi5sb2NhbDAqMAUGAytlcAMhADimJYKHpwsLe1eA
HPaChkVq34880OKo0UkZQWxRsfn5o0IwQDAdBgNVHQ4EFgQUTGezEi6EbU4MbrNz
V0fX1uTzh7swHwYDVR0jBBgwFoAU+NqNt20XF9rBmxb6rIWOW2QJn5cwBQYDK2Vw
A0EAZ9rDWVuNcax62B+e4OSQiFEUoFaLx4oE8krbpret7cokPfo5iOrIQA/ULzE2
zIWhs57hdtxvWFKYlZTkKmj+Dg==
-----END CERTIFICATE-----
PS C:\Users\shevc>
```

Certificate Details:

openssl x509 -in server.crt -text -noout

Certificate:

Data:

Version: 3 (0x2)

Serial Number:

6b:77:ec:ed:85:24:54:e4:c9:07:0b:98:ee:2d:1c:7e:f9:51:69:7c

Signature Algorithm: ED25519

Issuer: C=UA, ST=Some-State, O=Internet Widgits Pty Ltd, CN=Lab Root CA

Validity

Not Before: Apr 6 15:31:40 2026 GMT

Not After : Apr 6 15:31:40 2027 GMT

Subject: C=AU, ST=Some-State, O=Internet Widgits Pty Ltd, CN=secure-lab.local

Subject Public Key Info:

Public Key Algorithm: ED25519

ED25519 Public-Key:

pub:

38:a6:25:82:87:a7:0b:0b:7b:57:80:1c:f6:82:86:

45:6a:df:8f:3c:d0:e2:a8:d1:49:19:41:6c:51:b1:

f9:f9

X509v3 extensions:

X509v3 Subject Key Identifier:

4C:67:B3:12:2E:84:6D:4E:0C:6E:B3:73:57:47:D7:D6:E4:F3:87:BB

X509v3 Authority Key Identifier:

F8:DA:8D:B7:6D:17:17:DA:C1:9B:16:FA:AC:85:8E:5B:64:09:9F:97

Signature Algorithm: ED25519

Signature Value:

67:da:c3:59:5b:8d:71:ac:7a:d8:1f:9e:e0:e4:90:88:51:14:

a0:56:8b:c7:8a:04:f2:4a:db:a6:b7:ad:ed:ca:24:3d:fa:39:

88:ea:c8:40:0f:d4:2f:31:36:cc:85:a1:b3:9e:e1:76:dc:6f:

58:52:98:95:94:e4:2a:68:fe:0e

Verification result:

```
PS C:\Users\shevc> openssl verify -CAfile rootCA.crt server.crt
server.crt: OK
```

Assessment Questions:

Digital Signatures:

A **Man-in-the-Middle (MitM) attack** occurs when an attacker intercepts communication between a client and server and impersonates one or both parties. Without certificates, the client has no way to verify the server's identity, so it may unknowingly send sensitive data to the attacker.

A **Certificate Authority** prevents this by digitally signing certificates. The client trusts the CA and uses its public key to verify the certificate's signature. If the certificate is valid, the client knows it is communicating with the legitimate server, preventing impersonation.

TLS/SSL:

After verifying the certificate, the browser extracts the server's public key and uses it during the TLS handshake. Modern TLS uses **Elliptic Curve Diffie-Hellman Ephemeral**: In this process, both the browser and the server generate temporary pairs of private and public keys. They exchange these temporary public keys, and the server signs its exchange parameters with its long-term private key to prove its identity. Through the mathematical properties of elliptic curves, both parties can calculate a shared "Symmetric Session Key" without ever transmitting it over the network. This provides **Forward Secrecy**, ensuring that even if the server's long-term private key is stolen in the future, past recorded sessions remain encrypted and unreadable.

IPsec / VPN:

IPsec VPNs provide secure, encrypted tunnels for remote workers and branch offices to connect to the corporate network. While there are multiple ways to authenticate these connections, the choice between **Pre-Shared Keys (PSK)** and **Certificates** is a fundamental architectural decision.

Compromise Scenarios:

If server.key is stolen:

- Attacker can impersonate the server
- Can decrypt traffic (depending on setup)
- Impact is limited to that server

Severity: High, single system affected.

If rootCA.key is stolen:

- Attacker can issue fake certificates for any server
- Entire PKI trust is broken
- All clients trusting this CA are vulnerable

Severity: Critical, entire system compromised.